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Ethno-medico-botanical survey in Bishnugarh block of Hazaribagh, Jharkhand, India

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Abstract : Bishnugarh block, located in Hazaribagh District of Jharkhand State harbour rich plant diversity.The tribal communities, living in and around the sacred groves are endowed with rich traditional botanical knowledge pertaining to medicinal values of plant species. In the present study, we documented 20 vascular plant species of medicinal value used by the tribes. The paper deals with the systematic enumeration of the species with brief description, their distribution, phenology, local hindi names, and medicinal uses along with mode of administration. Over-exploitation and unscientific collection of some medicinal plants threatening the resource and warrants sustainable harvesting by the local communities.

Key words: Ethno- medico- botany, Sacred Groves, Bishnugarh, Hazaribagh District, Jharkhand.

INTRODUCTION

The **tribes of Jharkhand** consist of 32 tribes inhabiting the Jharkhand state in India. They are known as Munda, Santhal, Oraon, Kharia, Gond, Kol, Kanwar, Savar, Asur, Baiga, Banjara, Bathudi, Bedia, Binjhia, Birhor, Birjia, Chero, Chick-Baraik,Gorait, Ho, Karmali, Kharwar, Khond, Kisan, Kora, Korwa, Lohra, Mahli, Mal-Paharia, Parhaiya, Sauria-Paharia, Bhumij.These tribals have close association with and good knowledge about plant resources of their surroundings. With the reach of civilization to the traditional socities ,the traditional knowledge on the uses of these plants is fast vanishing. These are every need to document these knowledge, as otherwise it will be lost forever. So far there is no ethnobotancial study on the tribal of Bishnugarh block of Hazaribagh. Hence, this study was undertaken to record the traditional usuage of plants by the tribal people of this district.

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Study and area entnology

The Bishnugarh block is situated at 40 kms from district headquarter Hazaribagh. The term Hazaribagh s etyomological significance is 'A City of Thousand Gardens'. Occupying an area of 6147 sq. km and located at 23.98°N – 85.35 °E, Hazaribagh is 2019 ft. high above sea level in the Chottanagpur Plateau .

The main rivers in the district named as Damador ,Baraker, Lilajan and Mohani. Bishnugarh block of Hazaribagh district of Jharkhand area is very important because there are large number of tribals and indigenous persons residing in the heart of this district. This area holds importance because the life style, culture and customs of the inhabitants is not much affected by the modernization. They are still very close to the nature and have believe towards natural resources. They are still dependent on forest products for their livelihoods.

Bishnugarh is a beautiful hamlet situated 40 KMs from Hazaribagh town at its eastern side. Its beauty lies in its diverse landscapes, flowing rivers, dense forests and of course lovely populace. Geographical position of Bishnugarh block is situated North – East of Jharkhand. The block in district is district is spreaded over

27432.15(Acres) in area, consisting of forest, barren land (un- irrigated) and cultivated land. The population of the Bishnugarh block is above 1,42,862. The block consists of 28 panchayats.

There are four types of tribles as well as ethnic groups named as Shanthal, Ho, Oraon and Bhujan are using the medicinal and other plants. The tribals living in the regions speak their own dialects.

Bishnugarh block of is very rich in plants bio-diversity but unfortunately remained ignored on account of its floral study. It is inhabited by different kinds of ethnic groups. Majority of the populations is lying under poverty line and therefore deprived of medical and other facilities. They depend solely on ethnomedicine for the treatment of different diseases.

MATERIALS AND METHODS

Extensive ethno-medico-botanical survey was made in the rural and tribal pockets of Bishnugarh block of Hazaribagh district. The data presented are based on first hand information collected during the period from July 2012 to January, 2013. The information was recorded from tribal herbal medicine practitioners (many informants, *i.e.,* tribal doctors of 40 -70 years age) in different ethnic groups, through interviews, discussions, personal contacts and keep observations. Voucher specimens have been deposited in the herbarium of the botany Department, Ranchi University, Jharkhand, India.

RESULTS

Plants species are arranged alphabetically with their botanical name, family (in brackets),vernacular name(Hindi), medicine uses, dosage and mode of administration.

Aerva lanata (L.) Juss.(Amaranthaceae), Gorakhbuti or Kapuri jadi.Water extract (50ml) prepared from leaves (100 g) is given orally (10 ml) twice a day for 10days to cure urinary disorders.Traditionally by the tribal people, ethinic and vaidya, the plant is used in cough, strangury (slow to be and painful discharge of urine), headache and urolithiasis.The roots are used in the treatment of headache. The plant is regarded as a demulcent. It is valued for vermifuge for children. The ethinic give orally the juice of the roots to patients of liver congestion, jaundice, biliousness and dyspepsia. They also give decoction of the whole plant to cure pneumonia, typhoid and other prolonged fevers.Pharmacological studies reported diuretic[1], anti-inflammatory, hypoglycemic[2], anti-diabetic[3], antiparasitic[4], antimicrobial[5], hepoprotective[6], anti-urolithiasis[7],acute renal failure[8], antiasthmatic[9], antifertility[10] and hypolipidemic[11], immunomodulatory and antitumor[12,13], Anti-diarrheal[14] properties of *Aerva lanata*.

***Ammania baccifera* L. (**Lythraceae) Crape Myrtle family Agnibuti, **Dadamari, Kuranta** The tribal people, ethinic. Vaidya treat Ringworm with shoots of 5- 6 plants approx 3 -6 gms (the number may be more if the size of the lesion is big)Then the herb is ground with the seven fruits of maricha(local name golmaricha, Botanical name Piper nigrum L. family- Piperaceae) and made into a fine paste.The paste is applied on the lesion of Ringworm and kept for one hour to one hour. A single application is sufficient to cure the lesion. Some times one more application is required to cure the lesion. Leaves used to treat parasitic skin affection.[15] and diseases.[16] by them.

Abrus precatorius L.(Fabaceae), Ratti; Gaungchi; Gunchi; Gunja. Decoction prepared from fresh leaves (50g) with ginger (..0BZn gnber offncn ale) (5 g) is given orally one spoon per day for menstrual pains. A tea is made from the leaves and used to treat fevers, coughs and colds.[17] The roots, the stem and the leaves are effective in the treatment of coryza, cough, fever, jaundice resulting from viral hepatitis, and intoxications.[18] They are used as an edulcorating agent in composite recipes. The daily dose is 8 to 16g in the form of a decoction. The seeds, very toxic, are applied externally in an antiseptic and anti-inflammatory poultice to accelerate the bursting of boils and to cure mastitis and galactophoritis. The seeds contain a protein-based poison called abrin.[18] One seed contains enough poison to kill a human, but the seeds are very hard and must be chewed or ground to release the poison. However, cooking destroys the poison so that the seeds may be eaten.

Aristolochia bracteolata Lam. (Aristolochiaceae), Kitamari. Leaves (100 g) and 5g of black pepper (Pnper ngrum) are boiled with a glass of water. The decoction is taken for 21 days for chest pain.Whole plant used by the tribal people and ethinics for the treatment of Dermatitis, allergic disorder, leprosy, jaundice[19]-worms, fever, Leaves used for Anti inflammation[20], dermatitis, rashes[21], skin

disease, for scorpion sting[21,] Antipyretic, snake bite[22], Antiulcer, amenorrhoea, antihelmintic,[23] Antiplasmodial.[24,25] Seeds for Antibacterial, anti inflammatory and analgesics purposes. Roots for the treatment of Syphilis, gonorrhoea & skin diseases, eczema .[23]

Annona squamosa L. (Annonaceae), ata, sharifa, sitaphal. Seed paste is applied on head to kill lice. Powder of *Annona squamosa* roots and *Tamrindus Indica* leaves is given for reducing weight. Seed power of *Annona squamosa, Mucuna Pruiens* and *Abrus Precatorius* (white) is given to men for strength.Root paste is applied on the swelling of snakebite. A bark decoction of this plant is used to prevent diarrhea, while the root is used in the treatment of dysentery. A decoction of the leaves is used for cold and to clarify urine. Leaves are used to treat hysteria and fainting spells. The fruits of *Annona* are haematinic, cooling, sedative, stimulant, expectorant and maturant tonic. They are useful in treating anemia and burning sensation. The seeds are abortifacient and insecticidal and are useful in destroying lice in the hair. The bark and leaves contain annonaine, an alkaloid which is found to posses many of these properties.[26] Hypoglycemic and antidiabetic effect of *Annona squamosa* was reported in the leaf extract. From[27,28] the bark of *Annona squamosa*, a bioactive acetogenin with anticancer activity have been isolated.[29,30]

Asparagus racemosus Willd (Asparagaceae), Shatavari, Shatuli, Vrishya. Tubers (25g) are mixture with black pepper (25g) and the mixture (50:50) taken internally for stomach pain, one spoon twice a day for 7days. *A. racemosus* has also been used successfully by some Vaidya for nervous disorders.[31]The plant also has antioxidant, immunostimulant, anti dyspepsia and antitussive effects."[32] The roots are used for treatment by Vaidya, following a regimen of processing and drying. It is generally used as a uterine tonic, as a *galactogogue* (to improve breast milk), in hyperacidity, and as a general health tonic.

Boswelli serrata Roxb. (Burseraceae), Salai. Decoction prepared from stem bark (30 g) and turmeric(10 g)is taken internally for fevers, one spoon thrice a day for 15 days. Its ethno- medicinal use is for the treatment of rheumatoid arthritis and other inflammatory diseases.[33]

Centella asiatica (L.) Urban (Apiaceae), Bemgsag, Brahma-Manduki, Gotukola, Khulakhudi, Mandookaparni. Leaves (50 g) crushed with 5g Curcuma longa powder is applied externally for skin diseases.The major ethnomedicinal uses of the plant appears to be to alleviate gastrointestinal disorders likedysentery, constipation, stomach problems, indigestion and loss of appetite, and to enhance memory or to serveas nerve stimulant. gastrointestinal disorders, and stimulation of nerve or for treatment of mental retardation, treatment of headache, toothache, cuts andwounds, leucorrhea, skin disorders (like eczema, carbuncle), hemorrhoids, as antidote to poison, urinary troublesand leucorrhea, pneumonia, syphilis, liver problems (like jaundice), sexual weakness in males, fever, sun stroke,rickets, cardiovascular disorders, leprosy, tuberculosis, asthma, and varicocele.

Cascabela thevetia (L.) Lippold.(Apocynaceae), Peeli kaner. Latex (10ml) is topically applied for muscle pains. Latex of stem bark is applied on boils. Infusion of the bark is used in malaria. Seeds are used as antifertility agent. Leave ground and the paste plastered till cure in case of bone fracture. Mature seeds are used as an abortifacient and as purgative in rheumatism and dropsy. Fruit latex is applied in nail infection. Leaf paste mixed with castor oil and after little warming applied thrice daily for 2-3 days on external injuries as a pain alleviator.[34]

Canavalia gladiate (Jacq.) DC. (Fabaceae), **Lalkududumpal**. Leaf juice (10ml) is taken (one spoon thrice a day for 7days) to kill intestinal worms. Pod is used in the treatment of Burning, antiallergic, antiinflammatory, antiviral and antioxidant activities Anticancerous including anti carcinogenic properties.[35]

Canscora decussata Schult *et* Schult. F. (Gentianaceae), Sankhabati, Shankhaphuli, Shankhapuspi, Samkhaphuli. whole plant (50g) ground with honey (5ml) is applied (1ml) for mouth ulcers twice a day till cure. Traditional healers prescribe it as a brain tonic, nervine tonic, alternative, and laxative. antianxiety effects, memory invigorator, It is used in cerebral abnormalities, epilepsy, insomnia, burning sensation, edema, urinary disorders, snake-bites, and diseases caused by evil spirits. It is the best tonic for brain and nerves and was also recommended for sexual and seminal debilities.[36]

Capparis floribunda wight (capparaceae),) kiari, kobra. Fresh leaves (5-10) dipped in 10 ml castor oil (*Rnch*

*us communs)*are mildly heated and applied on boils blisters.

Cassytha filiformis L.(Lauraceae) akas bel. Whole plant (50g) crushed with turmeric (50g) is applied externally for body pain. Traditional remedy for cancer, plant is also used to treat various human birthing issues, gonorrhoea, kidney ailments , The plant possess activities such as anti-oxidant,anti-trypanosomal,anti-platelet, vasorelaxant activities.

Celastrus paniculatus willd (Celastruceae), Kondgaidh, Malkakni, Malkamni, Malkangni, Sankhu. Paste prepared from seeds (10 g) and tumeric (5 g)is gently massaged for body pains.Used for Appetizer, emetic, expectorant, sodorific,aphrodisiac, powerful brain and liver tonic,stimulate intellect and sharpen memory,thermogenic, emollient, nervine, stimulant,anti-inflammatory, diuretic, emmenagogue,diaphoretic, febrifuge, digestive, laxative,useful in abdominal disorders Seeds used externally on foul, indolent ulcers and scabies; useful both as an external and internal remedy in rheumatism, gout, paralysis, leprosy and weakness. Used against pruritis, skin diseases, leucoderma, cephalalgia, arthralgia, asthma, cardiac debility,inflammation, stranguary, neuropathy, amenorrhoea, dysmenorrhoea, epilepsy. Oil used for Intellect promoting, shows beneficial effect on the learning and memory process in mentally retarded children; thermogenic and rubefacient, stomachic, tonic, useful in abdominal disorders, enriches blood; powerful stimulant for neuromuscular system.[37]

Chloroxylon swietenia DC. (Flindersiaceae), Bhirra, Bhivia, Dhoura, Girya. 10g gum from stem dissolved in water (50ml) is taken (2 spoons per day ,for 10 da10 for uninary disorders, arthritis, rheumatism. In case of a problematic wound, the dried leaves are applied on wound in order to increase the healing process.

Cochlospermum religiosum (L.).Altson (Cochlospermaceae), Galgal . Paste stem bark (50g) is plastered over bone fracture belonging to the family *Cochlospermaceae* is a small deciduous tree.[38] The tree yields a gum which is known as katira. Traditionally it is used in treating cough, diarrhea, dysentery, pharyngitis, fistula, gonorrhea, trachoma and syphilis. The dried leaf and flowers are used as stimulants, antipyretic, laxative and sedative[39,40] Its leaves are traditionally used to treat jaundice and liver diseases[41] The root is found to be hepatoprotective.[42] Traditionally the bark is used for treating jaundice.[43]

Cissus quadrangularis L.(Vitaceae), commonly known as Hadjod. Stem is used for treatment of broken bones[44]. Mildly heated stem is applied externally for body pains[45], Very useful in reduced body weight, anti-inflammatory, Scurvy, Otorrhoea and epistaxis gynaecological disorders like menstrual problems[46]. Stem is effective in the treat ment of (Asthma, bites of poisionous insects, Burns and Wounds)[47,48,49] Sap with tamrind used to treat Gonorrea[49]. Antitumor potential is the exciting aspects of the plant. Several ethinic have reported the plant as a valuable antibacterial, antifungal and also active against other plant pathogens.

Cocculus hirsutus (L.) Diels (Menispermaceae), Patalagarudi, Chireta, Pharidbuti, Jalajamani, Chirhita. Leaf juice (50g) with honey (one spoon) is given twice a day for reducing excess heat due to fever. The root is bitter, acrid, alterative, laxative, demulcent, and antiperiodic in fever, tonic, and diuretic. The juice of leaves coagulates in water and forms mucilage, which is used externally as a cooling and soothing agent in prurigo, eczema, and impetigo[50,51] The roots of *C. hirsutus* have been mentioned to possess antiinflammatory and analgesic properties.[52]

***Mangifera indica* L.** (Anacardiaceae) *Amra, Chuta* Stem bark made into powder and taken internally twice a day for 3 days to cure *post delivery pain.* Leaves with the stem bark and seeds of *Ricinus communis* are boiled in water and the decoction is taken orally thrice in a day for3 days for *cough* and *asthma.* Resinous juice ismixed with the white of an egg and a teaspoonful of this mixture is taken orally thrice in a day for 3 days for *diarrhoea.*

***Physalis minima* L.(** Solanaceae**)** Used leaves by tribal healers, ethinics as *vulnerary;* leaves and ginger are ground well to get thick paste, gently applied on tumours, twice in a day until cured. Roots as *diuretic;* roots are cut into pieces, boiled in water to get decoction; a cup of decoction is taken in orally for 3 days. It is also used by them in the enlargement of spleen,gastric troubles, Fruit and flowers are used in stomach pain and in constipation. Herb paste is used in ear disorders. Considered as Antimalarial, Antigonorrheal, Antiinflammatory, Analgesic, Antipyretic, Antibacterial, Antiulcer, Antilipid peroxidation, Vermifuge, febrifuge, Abortificient.

DISCUSSION AND CONCLUSION

The present work is the result of intensive systematic ethnobotanical studies conducted in, Bishnugarh block of Hazaribagh district, India. In all 20 plant species of ethnobotanical interest are recorded after critical screening with the available literature and previous investigation if any about the concerned plant species.Uses of potential plant species are recorded. These are recommended for further rmore depth phytochemical/ Pharmacological investigation and nutritional analysis, which might result in the discovery of new drug molecules and also safe and effective administration of medicine for human welfare.

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